Retinoblastoma: proliferation up, retinal identity down

RNA-seq differential expression and pathway analysis of 19,952 genes

Swipe to see the results >>

The biological question

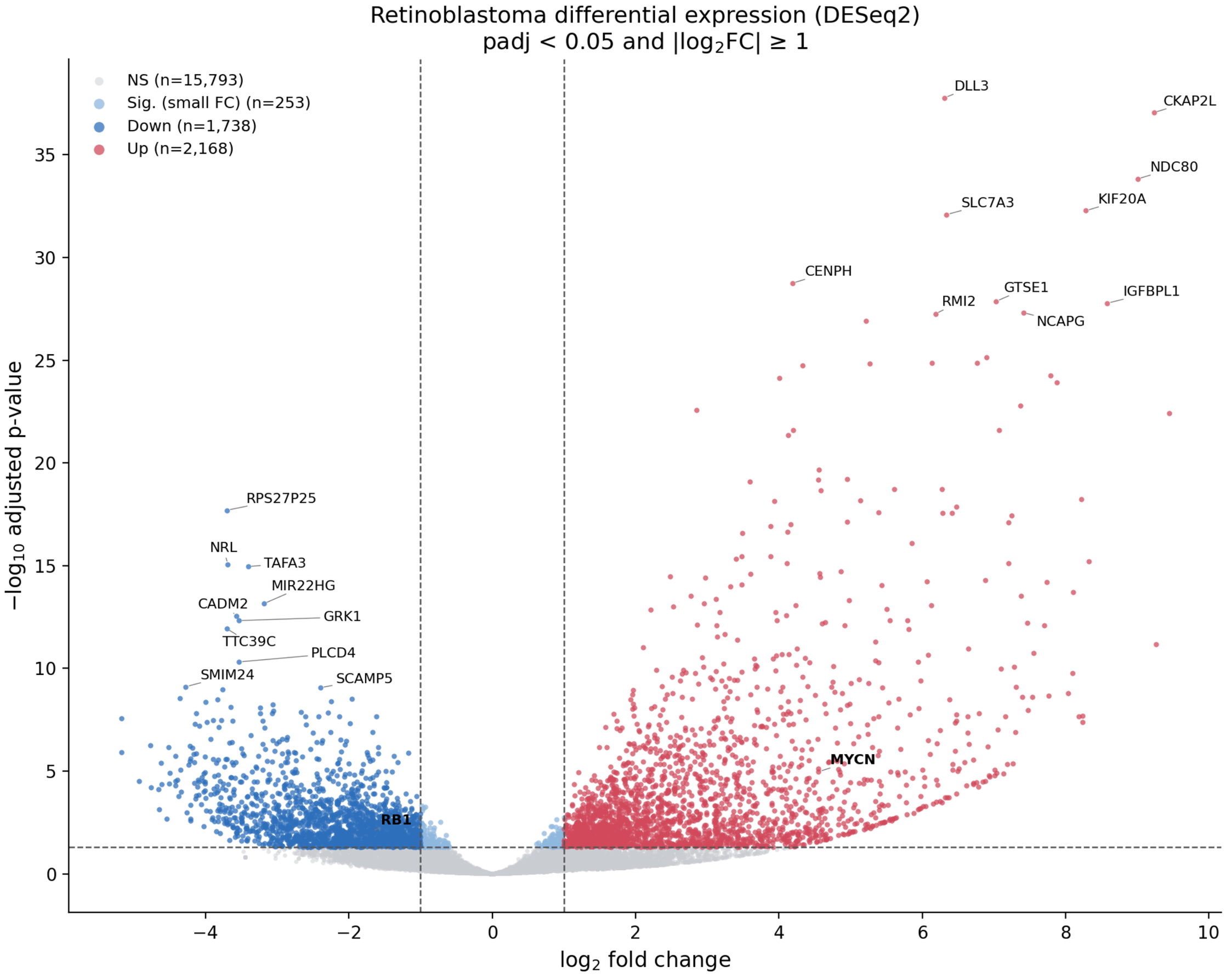
Retinoblastoma is a childhood eye cancer defined by loss of the **RB1** gene.

RB1 normally holds the **E2F** transcription factors in check. Without it, cell-cycle genes switch on.

Hypothesis

If the data are sound, cell-cycle and DNA-replication genes should be **up**, and retinal-specific genes should be **down**.

Volcano plot: 3,906 genes change strongly



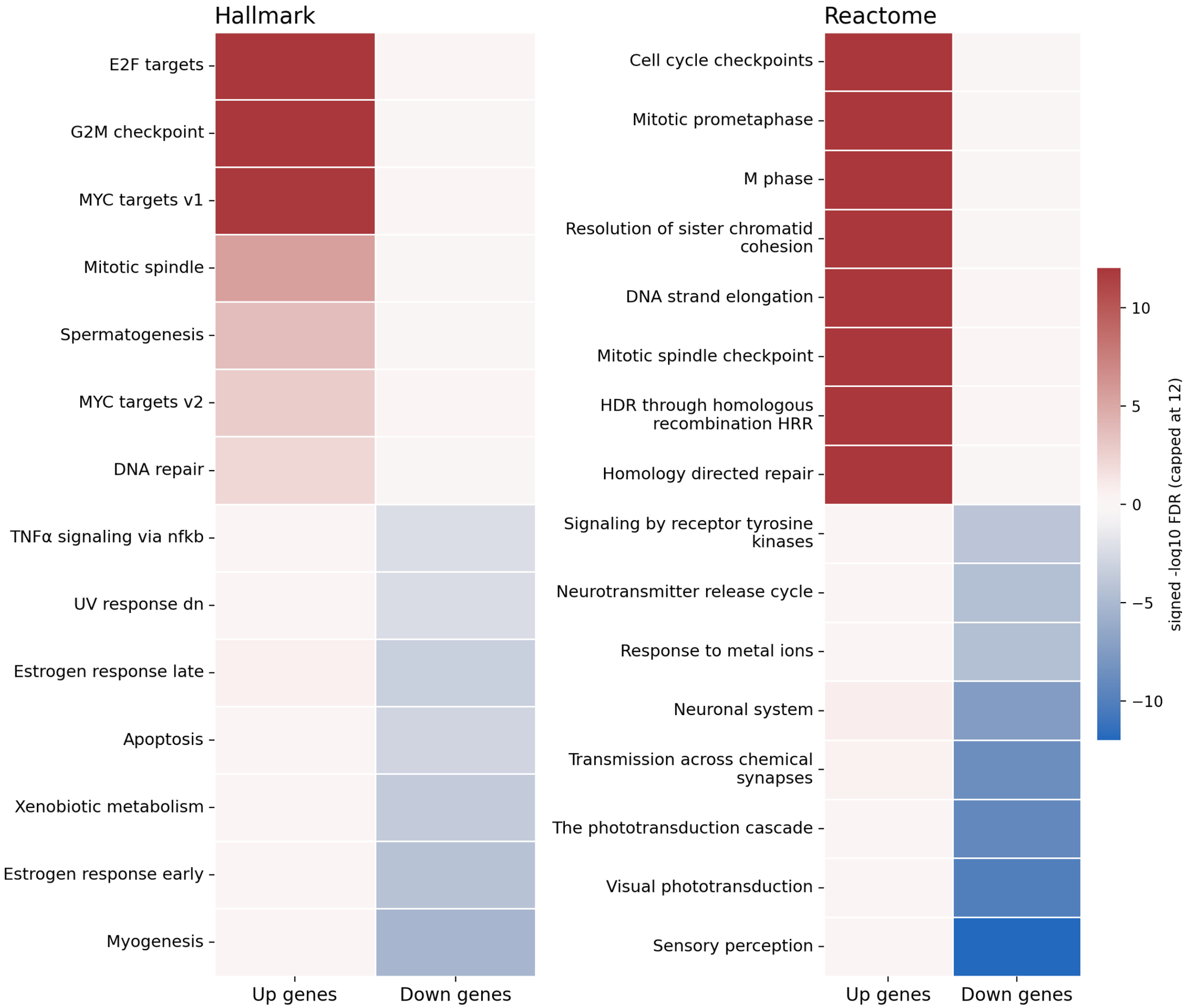
2,168 up and **1,738 down** (adjusted p < 0.05 and at least 2-fold change)

Known genes move in the expected direction

Gene	Role	log2FC	Adj. p	Direction
RB1	Tumour suppressor	-1.68	0.009	**Down**
MYCN	Oncogene	+4.55	1e-5	**Up**
MKI67	Proliferation marker	+2.77	0.031	**Up**
TOP2A	DNA replication	+3.93	0.0009	**Up**
NRL	Rod photoreceptor	-3.69	9e-16	**Down**
GRK1	Phototransduction	-3.54	5e-13	**Down**

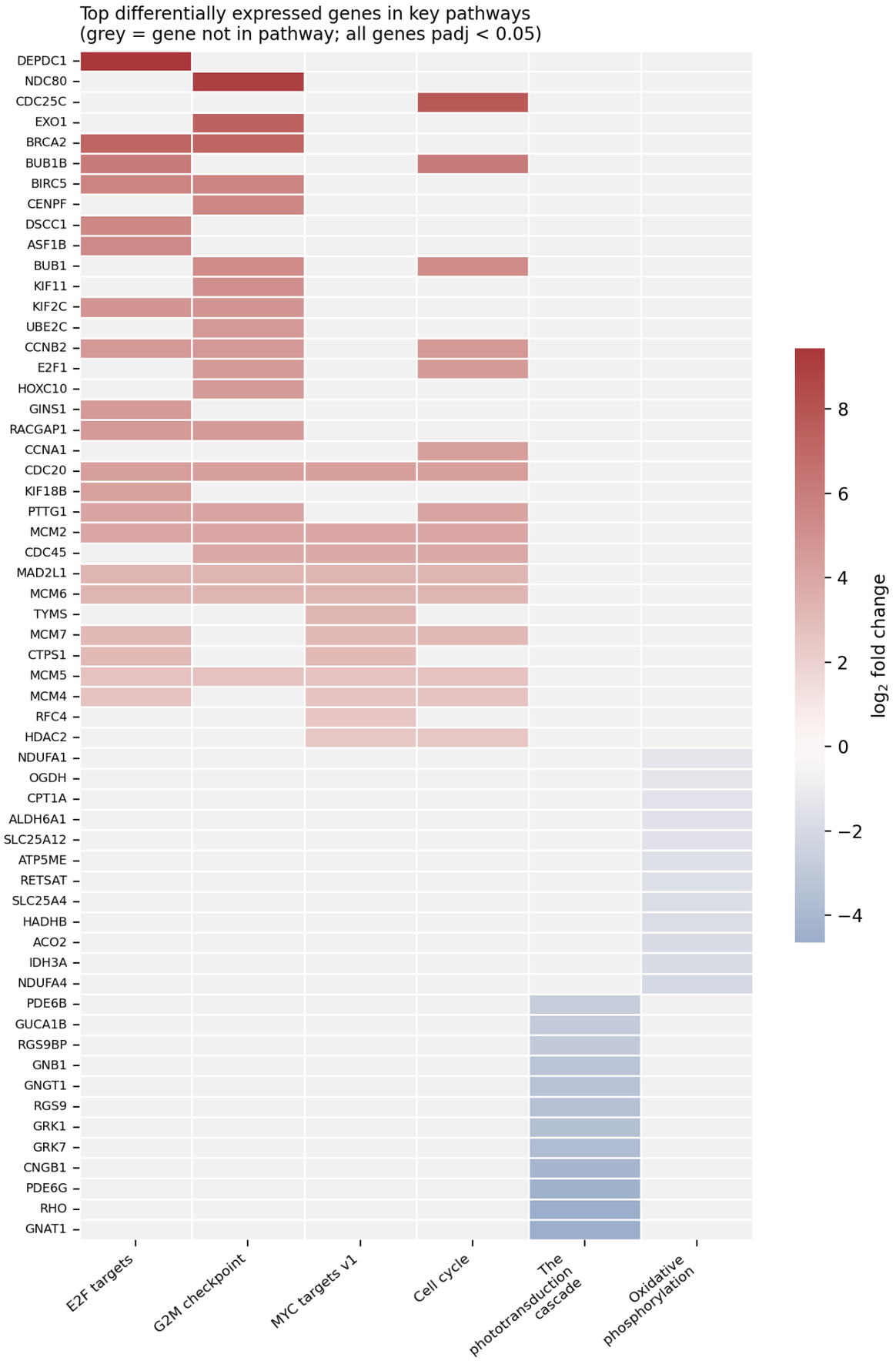
The tumour-suppressor is down, the oncogene and proliferation markers are up, and photoreceptor genes are down.

Two opposite programmes



Up: E2F targets, G2M checkpoint, MYC targets, DNA replication. **Down:** sensory perception and phototransduction.

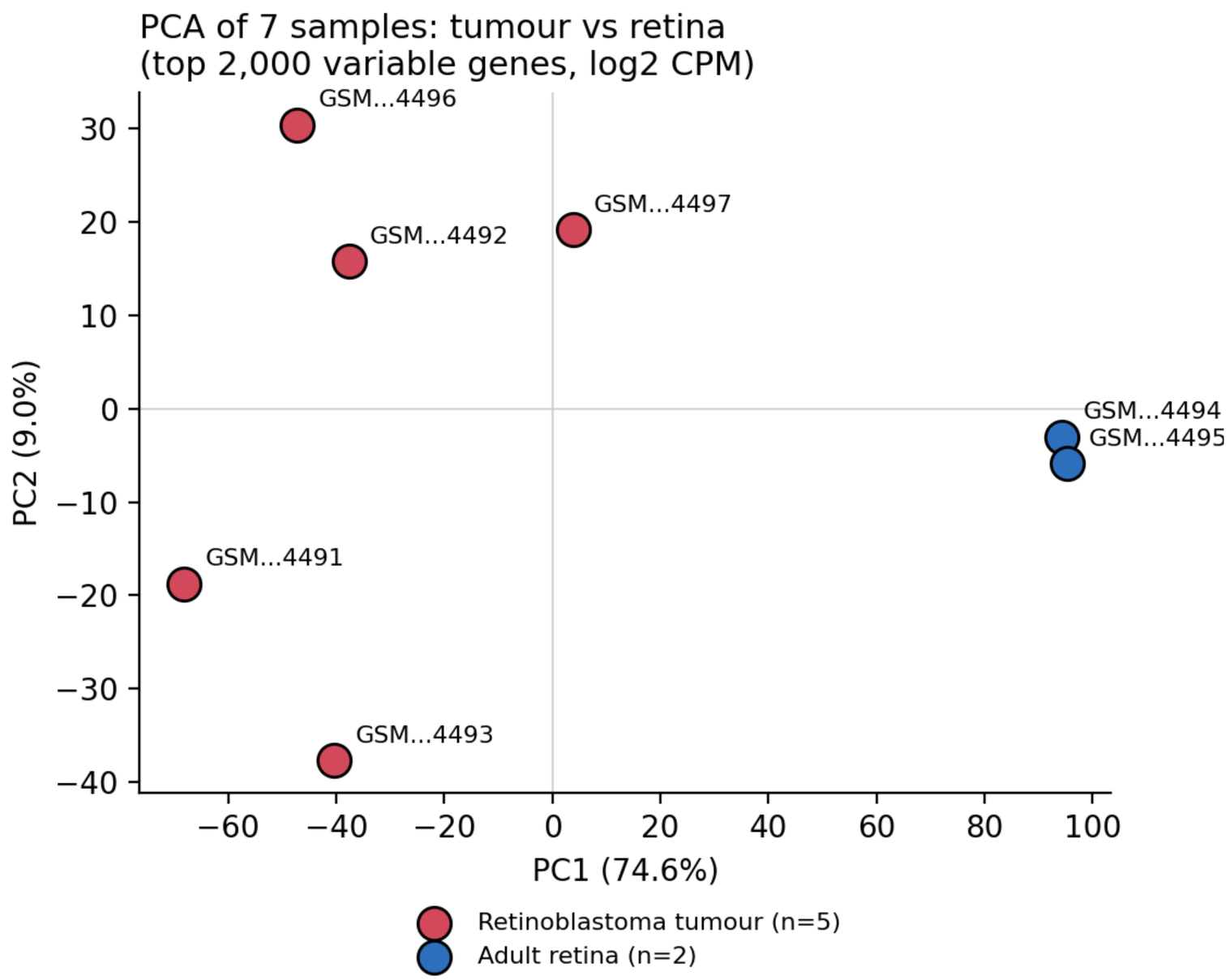
The genes behind the pathways



Up: NDC80, BUB1B, E2F1, MCM2 and MCM6 drive chromosome segregation and DNA replication.

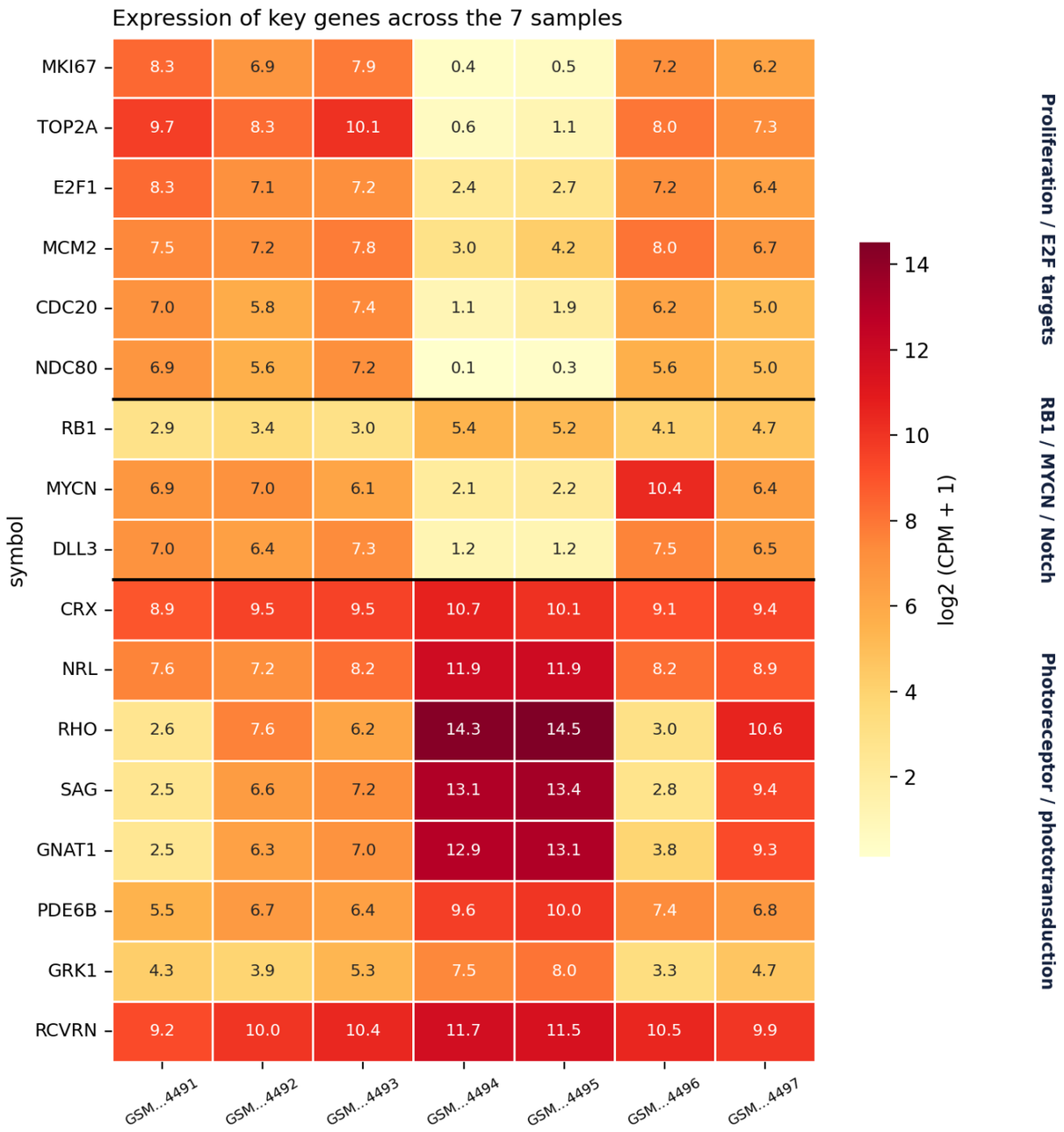
Down: RHO, GNAT1, PDE6B and GRK1 are core genes of the visual cycle in photoreceptors.

Tumours and retina separate cleanly



PC1 carries **74.6%** of the variance and splits the 5 retinoblastoma tumours from the 2 adult retina samples.

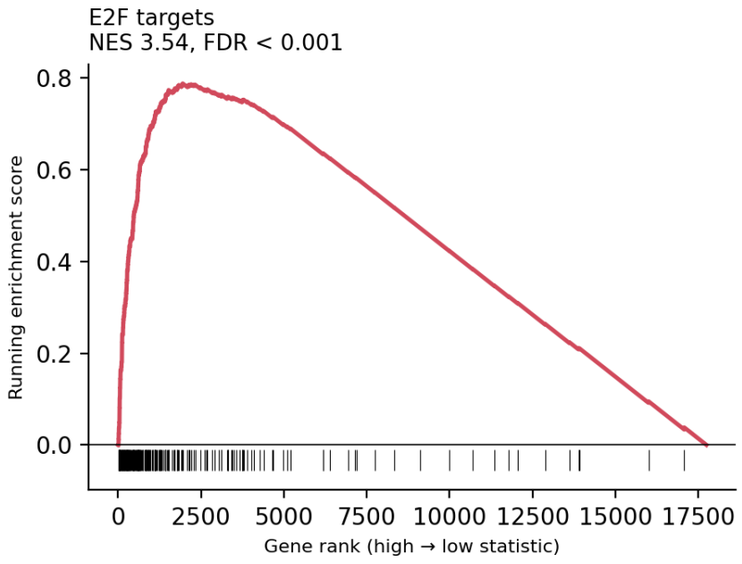
Each sample shows the expected genes

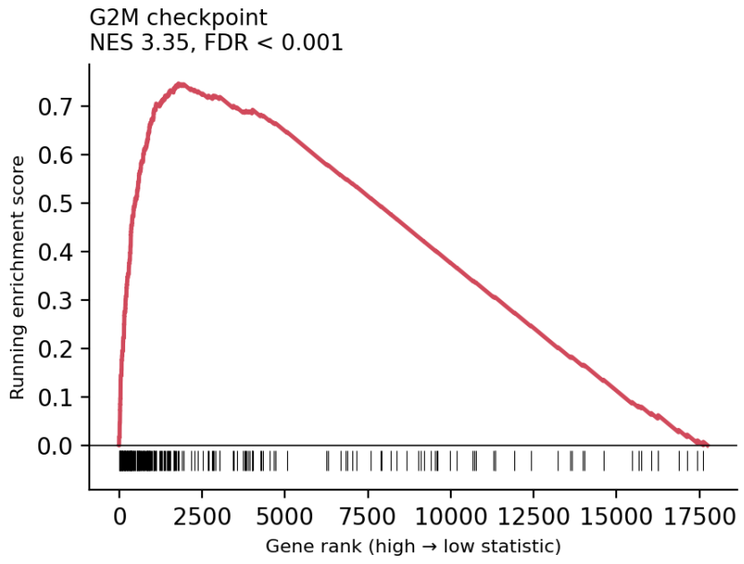


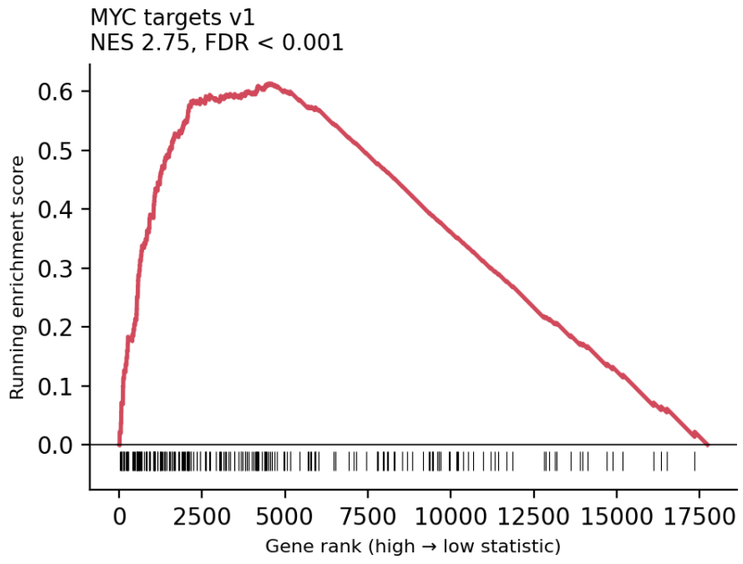
Tumours: proliferation genes high (MKI67, TOP2A, NDC80) and MYCN high.

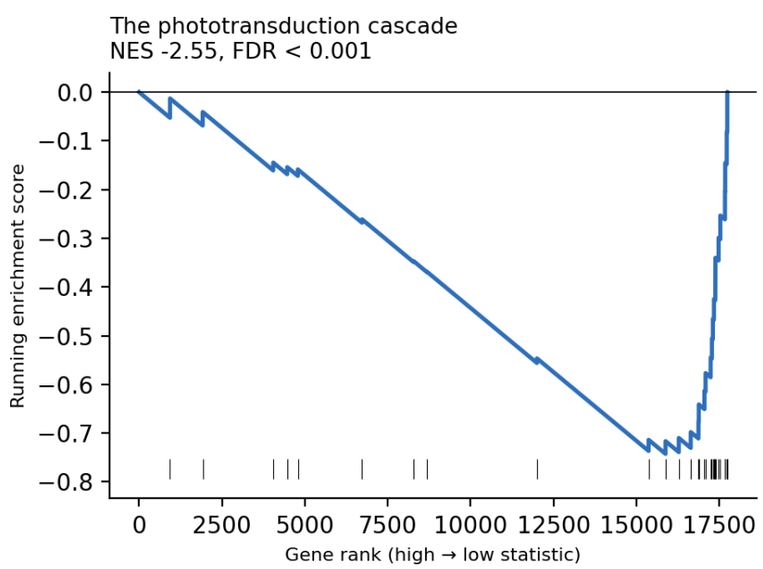
Retina: photoreceptor genes very high (RHO, SAG, GNAT1) and proliferation genes near zero. RB1 is higher in retina.

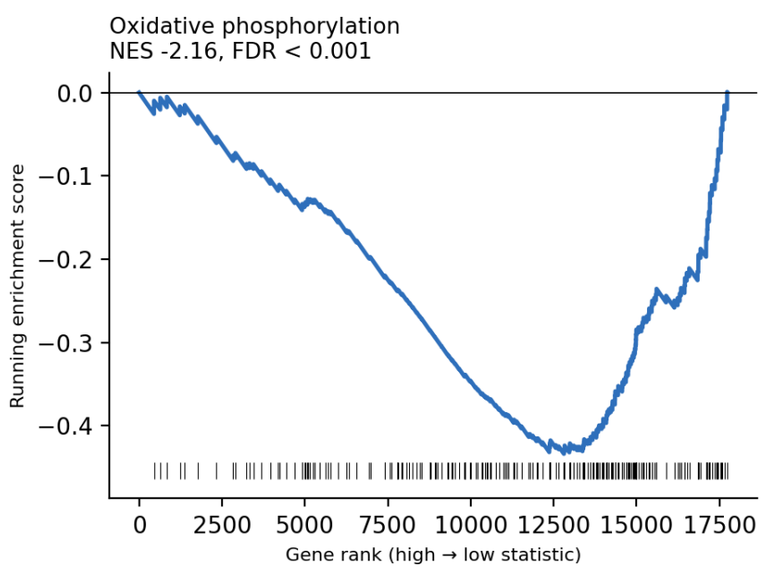
A second method agrees: GSEA

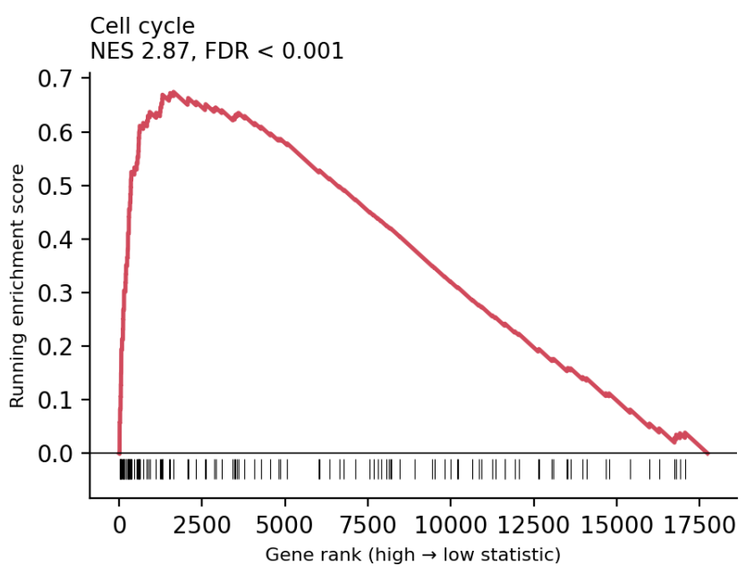












Preranked GSEA on all genes: E2F targets (NES 3.54), G2M checkpoint (3.35), phototransduction cascade (NES -2.55).

Querying the results with SQL

```
SELECT m.pathway,
       COUNT(*) AS genes_tested,
       SUM(g.status = 'Up') AS n_up,
       ROUND(AVG(g.log2fc), 2) AS mean_log2fc
FROM hallmark_membership m
JOIN genes g ON g.symbol = m.symbol
GROUP BY m.pathway HAVING COUNT(*) >= 30
ORDER BY mean_log2fc DESC;
```

Hallmark pathway	Genes	Up	Mean log2FC
E2F targets	198	135	+1.99
G2M checkpoint	198	116	+1.77
MYC targets V2	56	16	+0.94
MYC targets V1	199	60	+0.80

Limitations

Cell composition: the comparison group is adult retina, rich in photoreceptors, so the photoreceptor drop largely reflects tissue composition and stage, not only tumour changes.

Contrast direction: positive log2FC = higher in tumour (checked against the GEO counts, correlation 0.87).

Annotation: 11% of tested genes had no gene symbol and were excluded from enrichment.

Small groups: the counts file has 5 tumours and 2 retina samples (7 of 10 in the series).

Re-analysis: this recovers known biology and is not a new discovery.

How this was made

Data: public GEO dataset GSE125903 (retinoblastoma tumours vs adult retina), DESeq2 table of 57,773 genes.

Code: the Python analysis scripts and SQL queries were generated with AI assistance (Claude).

My contribution: framing the question, reviewing the outputs, checking them against known retinoblastoma biology, and interpreting the results.

Full code, SQL database and report on GitHub:
github.com/avani-0345/retinoblastoma-gene-expression-sql